Quadratic Equations

- What makes an equation quadratic
- One formula for every root
- How many roots, and what kind
- What the graph looks like

Standard Form

- ax^2 + bx + c = 0, with a not equal to 0
- a, b and c are the coefficients
- The solutions are called roots

Sridharacharya’s Formula

- x = (-b +/- sqrt(b^2 - 4ac)) / 2a
- Also called the quadratic formula
- Derived by completing the square

The Discriminant

- Delta = b^2 - 4ac
- Delta > 0: two real and distinct roots
- Delta = 0: real and equal roots
- Delta < 0: complex conjugate roots

Worked Example

- x^2 - 5x + 6 = 0
- a = 1, b = -5, c = 6
- Delta = 25 - 24 = 1
- x = 3 or x = 2

Vieta’s Formulas and the Parabola

- Sum of roots = -b/a
- Product of roots = c/a
- y = ax^2 + bx + c is a parabola
- Vertex at x = -b/2a